Next, we examine how to search two-dimensional (2D) shapes using *shape queries*. Design engineers, game developers, and others can search in an index of 2D shapes. Then we look at low-level positional queries called `span` *queries*. Although full-text and term-level leaf queries help us search for data, they can't find words in a particular order, their position, the exact (or approximate) distance between words, and so on. This is where `span` queries come into play.

Finally, we wrap up the chapter by looking at specialized queries such as `distance_feature`, `percolator`, `more_like_this`, and `pinned`. The `distance_feature` query boosts results that are nearer to a particular location: for example, searching for schools within a 10 km radius, but giving schools with nearby parks higher priority. To append organically found search results to a list of sponsored results, we use `pinned` queries. The `more_like_this` queries find similar-looking documents. The final specialized query we discuss is the `percolator` query, which helps notify users when data is available for queries that didn't yield results in the past.

Let's begin by discussing the need for geospatial queries and the data types that support them. We then look at the queries provided by Elasticsearch out of the box for such search criteria.

NOTE The code for this chapter is available on GitHub (http://mng.bz/2D6w) and on the book's website (https://www.manning.com/books/elasticsearch-in-action-second-edition).

NOTE Unlike other chapters, this chapter examines multiple types of queries including geospatial, shape, span, and specialized queries. Due to the nature of these queries, I prepared multiple data sets to satisfy their requirements. So, the examples shift between data sets (indexes) in this chapter.

12.1 Introducing location search

In this day and age of the internet, it is a common requirement to enable location-based search in apps and applications. Location-based search fetches venues or places based on proximity, such as nearby restaurants, houses for sale within a 1 km radius, and so on. We also use location-based searches to find directions to a place of interest.

The good news is that geospatial support is a first-class citizen in Elasticsearch. Dedicated data types allow us to define a schema for indexing geospatial data, thus enabling focused search. The out-of-the-box data types that support geospatial data are `geo_point` and `geo_shape`.

Elasticsearch also provides geospatial search queries that suffice for most use cases, such as `bounding_box`, `geo_distance`, and `geo_shape`. Each of these queries satisfies a set of requirements that we discuss briefly in this section and at greater length later in the chapter.

12.1.1 The bounding_box query

We sometimes want to find a list of locations such as restaurants, schools, or universities in a surrounding area—let's say in a square or rectangle. We can construct a